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Easy Level

① IP = 10.0.0.5/8
IP class = A
Default subnet mask = 255.0.0.0
Network bits = 8
Host bits = 24
Usable hosts per subnet = $2^{24} - 2 = 16777214$
Network address = 10.0.0.0
first usable IP = 10.0.0.1
last usable IP = 10.0.0.254
Broadcast address = 10.255.255.255

② IP = 172.16.5.10/16
IP class = B
Default subnet mask = 255.255.0.0
Network bits = 16
Host bits = 16
Usable hosts per subnet = $2^{16} - 2 = 65,534$
Network address = 172.16.0.0
First usable IP = 172.16.0.1
Last usable IP = 172.16.255.254
Broadcast address = 172.16.255.255

③ IP = 192.168.10.25/24
IP class = C
Default subnet mask = 255.255.255.0
Network bits = 24
Host bits = 8
usable hosts per subnet = 254
Network address = 192.168.10.0
First usable IP = 192.168.10.1
Last usable IP = 192.168.10.254
Broadcast address = 192.168.10.255

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(4) IP = 199.1.1.1/24
IP class = C
Default subnet mask = 255.255.255.0
Network bits = 24
Host bits = 8
Usable hosts per subnet = 254
Network address = 199.1.1.0
First usable IP = 199.1.1.1
Last usable IP = 199.1.1.254
Broadcast address = 199.1.1.255

(5) (i) IP = 15.10.5.20
IP class = A
Default subnet mask = 255.0.0.0

(ii) IP = 175.20.40.10
IP class = B
Default subnet mask = 255.255.0.0

(iii) IP = 210.100.5.1
IP class = C
Default subnet mask = 255.255.255.0

(6) IP = 192.168.1.100/24
Network address = 192.168.1.0
First usable IP = 192.168.1.1
Last usable IP = 192.168.1.254
Broadcast address = 192.168.1.255

(7) IP = 172.20.5.0/24
First usable IP = 172.20.5.1
Last usable IP = 172.20.5.254

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(8) IP = 5.1.2.3 ⇒ class - A
IP = 130.20.1.1 ⇒ class - B
IP = 192.168.1.1 ⇒ class - C
IP = 230.10.10.1 ⇒ class - D
IP = 250.1.1.1 ⇒ class - E

(9) IP = 192.168.50.10/25
Network address = 192.168.50.0
Broadcast address = 192.168.50.127
First usable IP = 192.168.50.1
Last usable IP = 192.168.50.126
Usable hosts Per subnet = 126

(10) IP = 10.20.30.40/16
Network address = 10.20.0.0
Broadcast Address = 10.20.255.255
First usable IP = 10.20.0.1
last usable IP = 10.20.255.254
usable hosts Per subnet = 65,534

Medium Level

(1) IP = 200.100.50.75/26
IP class = C
Default subnet mask = 255.255.255.0
subnet mask = 255.255.255.192
Network bits = 26
Host bits = 6
Network address = 200.100.50.64
First usable IP = 200.100.50.65
Last usable IP = 200.100.50.126
Broadcast address = 200.100.50.127

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(2) IP = 130.4.102.1/22
IP class = B
Default subnet mask = 255.255.0.0
subnet mask = 255.255.252.0
Network bits = 22
Host bits = 10
First usable IP = 130.4.100.1
Last usable IP = 130.4.103.254
Broadcast Address = 130.4.103.255
Network Address = 130.4.100.0

(3) IP = 203.1.1.0/28
subnet mask = 255.255.255.240
Number of subnets = 16
Usable hosts per subnet = 14
First four subnet blocks :-
① 203.1.1.0/28
② 203.1.1.16/28
③ 203.1.1.32/28
④ 203.1.1.48/28

(4) IP = 216.21.5.0/24
Borrowed bits = 3
New subnet mask = 255.255.255.224
usable hosts per subnet = 30
Number of subnets = 8
First two subnet blocks :-
① 216.21.5.0/27
② 216.21.5.32/27

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(5) IP = 192.168.100.0/26
Number of subnets = 4
Usable hosts per subnet = 62
First three subnet blocks :-
(1) 192.168.100.0/26
(2) 192.168.100.64/26
(3) 192.168.100.128/26

(6) IP-1 = 172.16.17.30/20 IP-2 = 172.16.28.15/20
Network = 172.16.16.0 Network = 172.16.16.0
Yes, Both belong to the same subnet.

(7) IP = 192.44.82.16/26
Network address = 192.44.82.0
Broadcast address = 182.44.82.63
Usable Range = 182.44.82.1 - 182.44.82.62

(8) IP-1 = 192.168.10.25/23 IP-2 = 192.168.11.200/23
Network = 192.168.10.0 Network = 192.168.10.0
Yes, Both belong to same subnet.

(9) IP = 172.16.10.100/20
subnet mask = 255.255.240.0
Usable hosts per subnet = 4094
Network address = 172.16.0.0
First usable IP = 172.16.0.1
Last usable IP = 172.16.15.254
Broadcast address = 172.16.15.255

(10) Need minimum 14 subnets
Borrowed bits = 4
New subnet mask = 255.255.255.240
Number of subnets created = 16
Usable hosts per subnet = 14

Hard Level

(1) IP = 10.10.0.0/16
Need ≥ 500 hosts
Minimum Prefix length = 23
subnet mask = 255.255.254.0
Borrowed bits = 7
Usable hosts per subnet = 510
Number of equal subnets = 128
First four subnet blocks :-
1. 10.10.0.0/23
2. 10.10.2.0/23
3. 10.10.4.0/23
4. 10.10.6.0/23

(2) IP = 172.20.0.0/16
Need ≥ 1000 hosts
Minimum Prefix length = 22
subnet mask = 255.255.252.0
Borrowed bits = 6
Number of subnets = 64
First four subnet blocks :-
1. 172.20.0.0
2. 172.20.4.0
3. 172.20.8.0
4. 172.20.12.0

(3) IP = 150.10.0.0/16
Needs 100 departments, 250 hosts each
subnet size = 256
Prefix = 24
subnet mask = 255.255.255.0
Available subnets = 256
Unused subnets = 156
usable subnets = 254

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First four subnet addresses :-
1. 150.10.0.0
2. 150.10.1.0
3. 150.10.2.0
4. 150.10.3.0

(4) IP = 172.31.100.0/21
Network = 172.31.96.0
Broadcast = 172.31.103.255
Usable range = 172.31.96.1 - 172.31.103.254
Usable hosts per subnet = 2046
No of /21 subnets from /16 = 32

(5) IP = 11.0.0.0/8
Need ≥ 2046 hosts
Prefix = /21
subnet mask = 255.255.248.0
Borrowed Bits = 13
No of subnets = 8192
First four subnet addresses :-
1. 11.0.0.0
2. 11.0.8.0
3. 11.0.16.0
4. 11.0.24.0

(6) IP = 100.10.0.0/8
Need ≥ 1200 hosts
Prefix = /21
subnet mask = 255.255.248.0
Borrowed Bits = 13
Hosts per subnet = 2046
No of subnets = 8192

First five subnet addresses :-

1. 100.0.0.0
2. 100.0.8.0
3. 100.0.16.0
4. 100.0.24.0
5. 100.0.32.0

(7) IP = 192.168.40.0.124
No of usable hosts to be supported 100, 50, 25.
method to be used : VLSI

Dept-1 100 Hosts
$2^7 - 2 = 126 \geq 100$ Prefix : /25
Network address : 192.168.40.0.125
Usable Range : 192.168.40.1 - 192.168.40.126
subnet mask : 255.255.255.128
Broadcast add : 192.168.40.127

Dept-2 50 Hosts
$2^6 - 2 = 62 \geq 50$ Prefix : /26
subnet mask = 255.255.255.192
Network address = 192.168.40.128
Usable Range : 192.168.40.129 - 192.168.40.190
Broadcast add : 192.168.40.191

Dept-3 25 Hosts
$2^5 - 2 = 30 > 25$ Prefix : /27
subnet mask : 255.255.255.224
Network address : 192.168.40.192
usable Range : 192.168.40.193 - 192.168.40.22
Broadcast add : 192.168.40.223

(8) IP = 172.16.0.0. /16
No of usable hosts to be supported : 400, 200, 100, 50.
method to be used : VLSM

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Dept-1 400 Hosts
$2^9 - 2 = 510 > 400$ Prefix: /23
Network add: 172.16.0.0/23
Usable range: 172.16.0.1 - 172.16.1.264
Broadcast address: 172.16.1.225

Dept-2 200 Hosts
$2^8 - 2 = 254 > 200$ Prefix: /24
subnet mask: 17
Network add: 172.16.2.0/24
Usable Range: 172.16.2.1 - 172.16.2.254
Broadcast add: 172.16.2.254

Dept-3 100 Hosts
$2^7 - 2 = 126 > 100$ Prefix: /25
Network address: 172.16.3.0./25
Usable Range: 172.16.3.1 - 172.16.3.126
Broadcast add: 172.16.3.127

Dept-4 50 Hosts
$2^6 - 2 = 62 \geq 50$ Prefix: /26
Network address: 172.16.3.128/26
Usable Range: 172.16.3.129 - 172.16.3.190
Broadcast Range: 172.16.3.191

(9) IP = 172.16.70.130/18
Network address: 172.16.64.0/18
Usable Range: 172.16.64.1 - 172.16.127.254
Broadcast address: 172.16.127.255

IP = 172.16.130.100/18
Network address: 172.16.128.0/18
Usable Range: 172.16.128.1 - 172.16.191.254
Broadcast address: 172.16.191.255

Hence, they do not belong in the same subnet.

(10.) IP = 192.168.1.0 /24
Least no of subnets needed : 6
Least usable hosts needed per subnet : 30

$2^3 = 8 > 6$
Borrowed bits : 3
New prefix : 24 + 3 = /27
Host bits : 32 - 27 = 5
Usable hosts : $2^5 - 2 = 30$
subnet mask : 255.255.255.254
No of subnet : $2^3 = 8$
Fixed length subnetting possible : yes

subnet block :
1. 192.168.1.0
2. 192.168.1.32
3. 192.168.1.64
4. 192.168.1.96
5. 192.168.1.128
6. 192.168.1.160
7. 192.168.1.192
8. 192.168.1.224